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Course : Data Structure and Algorithms I

Course Code: CSE 2215

Section : D

Assignment

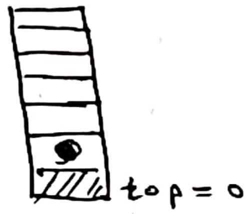
1)

(a) Using Array

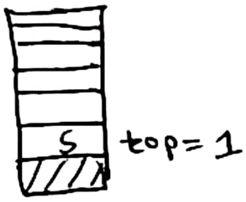
Assume Stack Size, $n = 6$

push (s), push (o), push(u), push(r), push (a), push(v)

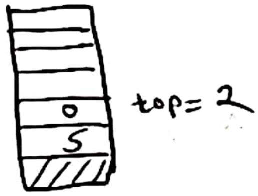
Initial:



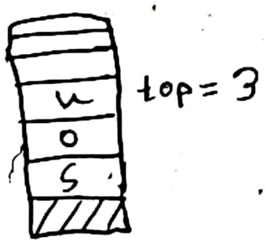
push (s)



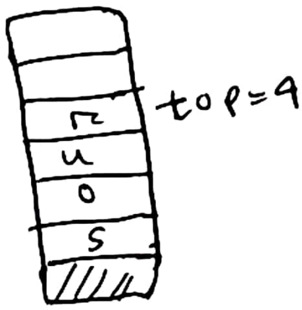
push (o)



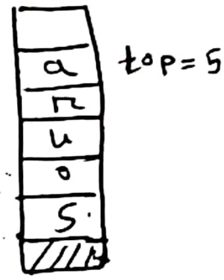
push (u)



push (r)

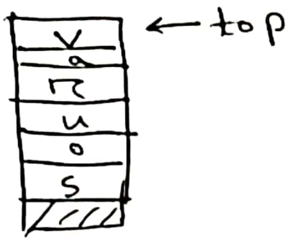


push (a)



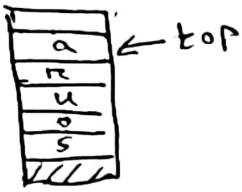
push (v)

Print (Top)



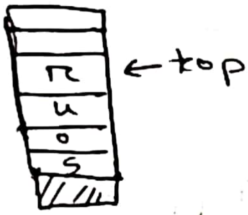
V

pop()



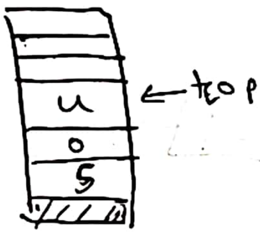
Va

pop()



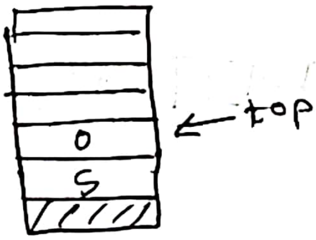
Var

pop()



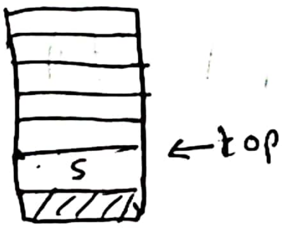
varu

pop()



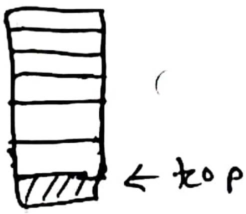
varuo

pop()



varuos

pop()



"Stack Empty"

1) (b) Using Linked List

Initial: top [////] NULL

push(s) [s |] → [////] NULL

top

push(o) [o |] → [s |] → [////] NULL

top

push(u) [u |] → [o |] → [s |] → [////] NULL

top

push(r) [r |] → [u |] → [o |] → [s |] → [////] NULL

top

push(a) [a |] → [r |] → [u |] → [o |] → [s |] → [////] NULL

top

push(v) [v |] → [a |] → [r |] → [u |] → [o |] → [s |] → [////] NULL

top

	Print (top) :
v → a → r → u → o → s → /// NULL (top = v)	V
pop(): a → r → u → o → s → /// NULL (top = a)	va
pop(): r → u → o → s → /// NULL (top = r)	var
pop(): u → o → s → /// NULL (top = u)	varu
pop(): o → s → /// NULL (top = o)	varuo
pop(): s → /// NULL (top = s)	varuos
pop(): /// → NULL (top)	"Stack Empty"

2) i

(a)

Infix Expression : $a \uparrow 2 - (b + c / d * a + b) + d$

Infix Expression	Stack	Postfix Expression
$a \uparrow 2 - (b + c / d * a + b) + d$		a
$\uparrow 2 - (b + c / d * a + b) + d$		
$2 - (b + c / d * a + b) + d$	$\uparrow$	a
$- (b + c / d * a + b) + d$	$\uparrow$	$a 2$
$(b + c / d * a + b) + d$	$-$	$a 2 \uparrow$
$b + c / d * a + b) + d$	$-($	$a 2 \uparrow$
$+ c / d * a + b) + d$	$-($	$a 2 \uparrow b$
$c / d * a + b) + d$	$-(+$	$a 2 \uparrow b$
$/ d * a + b) + d$	$-(+$	$a 2 \uparrow b c$
$d * a + b) + d$	$-(+/$	$a 2 \uparrow b c$
$* a + b) + d$	$-(+/$	$a 2 \uparrow b c d$

Infix Expression	Stack	Postfix Expression
a +b) + d	– (+ *	a2↑bcd/
+b)+d	– (+ *	a2↑bcd/a
b) + d	– (+	a2↑bcd/a*+
) +d	–(+	a2↑bcd/a*+b
+d	–	a2↑bcd/a*+b+
d	+	a2↑bcd/a*+b+–
	+	a2↑bcd/a*+b+–d
		a2↑bcd/a*+b+–d+

Final postfix Expression : a2↑bcd/a*+b+–d+

2)
(c) $a = 9$; $b = 10$; $c = 3$; $d = 1$

Postfix Expression	Stack	Operation
a2↑bcd/a*+b+-d+		
2↑bcd/a*+b+-d+	a	
↑bcd/a*+b+-d+	a2	
bcd/a*+b+-d+	81	$a \uparrow 2 = 9 \uparrow 2 = 81$
cd/a*+b+-d+	81b	
d/a*+b+-d+	81bc	
/a*+b+-d+	81bcd	
a*+b+-d+	81b3	$c/d = 3/1 = 3$
*+b+-d+	81b3a	
+b+-d+	81b27	$3 * a = 3 \times 9 = 27$
b+-d+	81 37	$b + 27 = 10 + 27 = 37$
+-d+	81 37 b	
-d+	81 47	$37 + b = 37 + 10 = 47$

postfix Expression	stack	operation
d+	34	$81 - \cancel{8}47 = 34$
+	34 d	
	35	$34 + d = 34 + 1 = 35$

Final Answer = 35.